Methods: Evidence Roles, Curriculum Graph, and Stratum Design

Source-role matrix for citation coverage and source health

Source-role matrix for citation coverage and source health I

The manuscript uses a second, non-numeric integrity gate for citations. The static `data/scholarship_sources.yaml` file maps each cited BibTeX key to a scholarship lane, source kind, locator, and support role. `src/scholarship/coverage.py` parses the authored manuscript sections, validates those keys against `manuscript/references.bib`, and rejects any cited key that lacks a matrix entry or authoritative locator. It also rejects stale source-matrix rows and uncited BibTeX entries, because those orphans would otherwise let the coverage artifact imply support that no manuscript sentence actually uses. The resulting JSON and figure are generated artifacts, not hand-maintained summaries. The source-health check adds a second layer over this matrix: DOI and URL locators must be syntactically authoritative, documentation/software rows must point to URLs, and each role must explain how the source supports the manuscript

rather than merely naming the source.

This gate does not decide whether a paper is true. It prevents weaker failure modes that ordinary citation checks miss: an undefined BibTeX key, a citation with no stated role, a new source absent from the coverage matrix, or a source lane that disappears from the manuscript without anyone noticing. The scholarship figure is therefore an audit view over the current manuscript, not a bibliography inventory.

The matrix is also a situation map for the paper's evidence boundary. Foundational and historical sources explain vocabulary and lineage; Active-Inference sources define or compare model formulations; critique and empirical-status sources calibrate generality and application language; pedagogy and assessment sources motivate ordering and future study design; platform and software sources establish delivery and implementation context; and provenance, reproducibility, accessibility,

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and security sources describe artifact obligations. A source can occupy more than one lane, but its role is declared at the claim boundary rather than inferred from prestige, recency, or citation count.

Curriculum graph model, DAG validation, and quality gates

Pedagogical quality constraints I

`src/curriculum` defines typed `Stratum`, `Subject`, `Skill`, and `AssessmentQuestion` records, with field names mirroring SkillTree so export is a thin map. The `CurriculumGraph` validates structural invariants – unique identifiers, no self-loops, no dangling prerequisites, no cycles, and roots only in the two foundational strata – and provides deterministic topological order, per-skill depth levels, and a reachability check from foundations to applications. Cycle detection uses the standard three-color depth-first-search pattern from graph algorithms (Cormen et al. 2022).

Pedagogical quality constraints I

Structural validity is necessary but not sufficient for learning quality, so the build also runs `src/curriculum/quality.py`. That gate rejects shallow one-question skills, missing tags, terse subject or skill descriptions, answer-position bias, and backward prerequisites that would force later-stratum concepts into earlier learning paths. Such cross-stratum references are retained as non-blocking related links rather than exported as SkillTree dependencies. A semantic-prerequisite gate additionally rejects any skill that is *assessed on* a load-bearing concept (KL divergence, Markov chains, the Laplace approximation, the ELBO) without the concept's defining skill in its transitive prerequisites, flagging pedagogical inversions while exempting forward references for motivation; a non-empty-scan guard fails the build if any guarded concept stops appearing in the live corpus, so the gate cannot silently become vacuous.

Pedagogical quality constraints I

The newest hardening layer expands continuous dynamics, state-space probability, rate-distortion information, continuous variational inference, predictive-coding process theory, hierarchical active inference, reproducible simulation, and applied evaluation dossiers. Those additions are grounded in filtering, information bottleneck, Helmholtz-machine, and generalized-filtering references while remaining pedagogical rather than production inference engines (Särkkä 2013; Tishby et al. 1999; Dayan et al. 1995; Friston et al. 2010).

Pedagogical quality constraints I

The same graph is educationally meaningful only in a narrower sense: it approximates a knowledge-space style closure condition over authored skills, where later tasks should not be treated as feasible before their prerequisites, but it does not infer an individual learner's latent knowledge state (Doignon and Falmagne 1985; Falmagne and Doignon 2011).

Stratum order as an inspectable historical and pedagogical design choice

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The eight-stratum ordering – mathematics, probability, information theory, variational methods, the Free Energy Principle, Active Inference, computation, and applications – is an authored design decision, not a unique decomposition of the domain. An applications-first path or a thematic split between generative-model statics and inference/action dynamics could serve different learners. This release chooses the linear order because the graph can then enforce a simple local rule: concepts precede the concepts that depend on them (Bloom 1956; Bloom 1968; Liang et al. 2018). Historical sources motivate the categories and expose older problems of demonstration, perception, probability, and ordered instruction, but they do not determine the graph edges or certify the selected order as historically inevitable (Aristotle 1901; Ibn al-Haytham 1989; Quintilian 95 AD; Huygens 1657).

Stratum order as an inspectable historical and pedagogical design choice I

Historical pedagogy sources support this as a bounded history of design concerns rather than as an efficacy claim: Bacon emphasizes staged inquiry, Comenius ordered progression, Locke habit and practice, and Rousseau developmental sequencing (Bacon 1620; Comenius 1657; Locke 1693; Rousseau 1762). Knowledge-space and learning-space theory provide a formal comparison for prerequisite-constrained mastery states, while curriculum-aware cognitive diagnosis work shows that a stronger learner-model claim would require response data and an explicit model of learner state (Doignon and Falmagne 1985; Falmagne and Doignon 2011; Fu and Fang 2025). Hugo's Active Inference account of teacher development, embodied cognition, and education-oriented Active Inference work motivate keeping the order revisable and uncertainty-sensitive, but this release does not evaluate teacher development, infer learner states, or

implement an adaptive diagnostic engine (Hugo 2026; Clark 2016; Di Paolo et al. 2024).

Microlearning and gamification sources motivate modular delivery while warning that implementation context, satisfaction, and measured effectiveness can diverge (Septiani and Rosmansyah 2021; Monib et al. 2025; Rof et al. 2024). Active-learning, epistemic-curiosity, cognitive-engagement, retrieval-practice, and evidence-centered-assessment sources therefore inform the design of interaction and quiz tasks; they do not substitute for data from this curriculum or establish durable competence (Freeman et al. 2014; Emin et al. 2026; Chi and Wylie 2014; Roediger and Karpicke 2006; Mislevy et al. 2003). Constructive alignment, mastery learning, formative assessment, and feedback provide the local rule that objectives, activities, and quiz gates should be co-specified and used as revision checkpoints (Biggs 1996; Bloom 1968; Black and Wiliam 1998; Shute 2008). The

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resulting ordering is a parameter to inspect and revise, not a theoretical or empirically tested claim about how the field must be structured.